

Olivia Bakke, Jan 24, 2026

The diagram illustrates the memory layout of a program, divided into three main sections: **globals**, **stack**, and **heap**. Above these sections, three status indicators are displayed: **line number 1**, **step number 1**, and **total steps 21**.

- globals**: This section is located at the bottom of the memory layout, starting at address 0x00000000 and ending at 0x00000004. It contains a single entry for the variable `global` at address 0x00000000, with a value of 0x00000000.
- stack**: This section is located in the middle of the memory layout, starting at address 0x00000004 and ending at 0x00000008. It contains a single entry for the variable `stack` at address 0x00000004, with a value of 0x00000000.
- heap**: This section is located at the top of the memory layout, starting at address 0x00000008 and ending at 0x0000000C. It contains a single entry for the variable `heap` at address 0x00000008, with a value of 0x00000000.

The memory layout is represented by a large rectangle divided into three horizontal sections. The **globals** section is at the bottom, the **stack** section is in the middle, and the **heap** section is at the top. Each section contains a list of memory addresses and their corresponding values. The **globals** section has one entry, the **stack** section has one entry, and the **heap** section has one entry. The values are all 0x00000000.

line number 2

step number 2

total steps 21

globals

stack

heap

id1

list

0 = id2 

1 = id3 

id2

list

0 = 2 

1 = 3 

id3

list

0 = 1 

1 = 1 

line number 3

step number 3

total steps 21

globals

row = id2 

stack

heap

id1 list 

0 = id2 

1 = id3 

id2 list 

0 = 2 

1 = 3 

id3 list 

0 = 1 

1 = 1 

line number 4

step number 4

total steps 21

globals

row

 = id2

num

 = 2

stack

heap

id1

list

0

 = id2

1

 = id3

id2

list

0

 = 2

1

 = 3

id3

list

0

 = 1

1

 = 1

line number 3

step number 5

total steps 21

globals

row

 = id2

num

 = 2

stack

heap

id1

list

0

 = id2

1

 = id3

id2

list

0

 = 4

1

 = 3

id3

list

0

 = 1

1

 = 1

line number 4

step number 6

total steps 21

globals

row

 = id2

num

 = 3

stack

heap

id1

list

0

 = id2

1

 = id3

id2

list

0

 = 4

1

 = 3

id3

list

0

 = 1

1

 = 1

line number 3

step number 7

total steps 21

globals

row

 = id2 

num

 = 3 

stack

heap

id1

list



0

 = id2 

1

 = id3 

id2

list



0

 = 7 

1

 = 3 

id3

list



0

 = 1 

1

 = 1 

line number

5

step number

8

total steps

21

globals

table

=

id1



row

=

id2



num

=

3



stack

heap

id1

list



0

=

id2



1

=

id3



id2

list



0

=

7



1

=

3



id3

list



0

=

1



1

=

1



line number

2

step number

9

total steps

21

globals

table

=

id1



row

=

id3



num

=

3



stack

heap

id1

list



0

=

id2



1

=

id3



id2

list



0

=

7



1

=

3



id3

list



0

=

1



1

=

2



line number 3

step number 10

total steps 21

globals

stack

heap

id1 list

0 = id2

1 = id3

id2 list

0 = 7

1 = 3

id3 list

0 = 1

1 = 2

line number 4

step number 11

total steps 21

globals

row

 = id3

num

 = 1

stack

heap

id1

list

0

 = id2

1

 = id3

id2

list

0

 = 7

1

 = 3

id3

list

0

 = 1

1

 = 2

line number 3

step number 12

total steps 21

globals

table = id1

row = id3

num = 2

stack

heap

id1 list

0 = id2

1 = id3

id2 list

0 = 8

1 = 3

id3 list

0 = 1

1 = 2

line number 4

step number 13

total steps 21

globals

row

 = id3 

num

 = 2 

stack

heap

id1

list



0

 = id2 

1

 = id3 

id2

list



0

 = 8 

1

 = 3 

id3

list



0

 = 1 

1

 = 2 

line number 3

step number 14

total steps 21

globals

table = id1

row = id3

num = 2

stack

heap

id1 list

0 = id2

1 = id3

id2 list

0 = 10

1 = 3

id3 list

0 = 1

1 = 2

line number 5

step number 15

total steps 21

globals

table = id1

row = id3

num = 2

stack

heap

id1 list

0 = id2

1 = id3

id2 list

0 = 10

1 = 3

id3 list

0 = 1

1 = 2

line number

2

step number

16

total steps

21

globals

table

=

id1



row

=

id3



num

=

2



stack

heap

id1

list



0

=

id2



1

=

id3



id2

list



0

=

10



1

=

3



id3

list



0

=

1



1

=

3



line number 6

step number 17

total steps 21

globals

table = id1

row = id3

num = 2

stack

heap

id1 list

0 = id2

1 = id3

id2 list

0 = 10

1 = 3

id3 list

0 = 1

1 = 3

line number 7

step number 18

total steps 21

globals

table = id1

row = id3

num = 2

new_row = id4

stack

heap

id1 list

0 = id2

1 = id3

id2 list

0 = 10

1 = 3

id3 list

0 = 1

1 = 3

id4 list

line number 8

step number 19

total steps 21

globals

table = id1

row = id3

num = 2

new_row = id4

stack

heap

id1 list

0 = id2

1 = id3

2 = id4

id2 list

0 = 10

1 = 3

id3 list

0 = 1

1 = 3

line number 9

step number 20

total steps 21

globals

table = id1

row = id3

num = 2

new_row = id4

first = 20

stack

heap

id1 list

0 = id2

1 = id3

2 = id4

id2 list

0 = 10

1 = 3

id3 list

0 = 1

1 = 3

line number 9

step number 21

total steps 21

globals

table = id1

row = id3

num = 2

new_row = id4

first = 20

second = 1

stack

heap

id1 list

0 = id2

1 = id3

2 = id4

id2 list

0 = 10

1 = 3

id3 list

0 = 1

1 = 3